



GETTING REALISTIC FAILURE RATE DATA

White Paper

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Abstract

This paper describes sources of failure rate data suitable for use of functional safety design verification. The two primary sources are classified as estimation and prediction. For those who use failure rate data as intended by our functional safety standards to verify design sufficiency and to guide optimal economic payback, **realistic data is necessary**.

Conclusion

Many methods have been defined for getting failure rate data. Some methods are motivated by a goal to show a manufacturer/industry superiority. Some methods created for one application/industry are incorrectly used in another. What is needed for functional safety is realistic data. Realistic failure rate data can be obtained from quality field failure data using a statistical estimation technique after a time period when data is gathered. Realistic failure rate data can also be obtained from prediction methods calibrated to the environment and application. The best approach is to use a combination of estimation and prediction. A method for calibrating an FMEDA component database is described which uses both estimation and prediction. Comparisons between predictions and estimates show that these two diverse methods produce similar results with realistic assumptions and a good component failure database.

Introduction - Failure Rate

The disciplines of Reliability and Functional Safety Engineering depend upon a key metric, failure rate. Two fundamentally different methods are used obtain these numbers, estimation and prediction.

Failure Rate Estimation

In the past it was practical to perform a “life test” (or an “accelerated life test”) where a set of devices were operated in a defined set of conditions and failure times were recorded. This failure time data was analyzed to obtain the failure rate of the device [1, 2]. This technique was far more practical with the technologies of 1950s and 1960s. Vacuum tubes failed so frequently that “tube testers” (Figure 1, provided assuming some readers have never seen a vacuum tube) were commonly seen in retail stores where new vacuum tubes were sold.



Figure 1 Consumer operated vacuum tube tester and vacuum tubes.

Today's technology is far more reliable for both electronic and mechanical devices such that the life test approach is rarely seen except perhaps for wear-out testing of mechanical devices [3].

In the 1950s failure reports from the field were also used to obtain an estimate of a device failure rate. A point estimate of the average failure rate was calculated by taking the failure count and dividing by an estimate of operational hours. This technique could be useful if the failure count and the estimated operational hours were accurate but **optimistic assumptions were often made and create a substantial error**. These assumptions have resulted in several orders of magnitude difference in the results [4, 5]. Consider the chart presented in a report from a manufacturer of electronic devices in Figure 2.

Table 1 - Reliability of Individual Parts					
Part Number	Description	Field Proven MTBF (Yrs)	Lambda FITS	Predicted MTBF	Lambda FITS
xxx	Analog Input Module	434.04	263	8.05	14181
xxxa	ROHS Analog Input Module	69.03	1654	7.99	14287
xyy	Analog Output w HART	570.91	200	9.81	11637
xxz	Fiber Optic Module	159.58	715	56.75	2012
xxa	Logic Solver	92.28	1237	7.38	15468
xxb	Logic Solver Fiber Optic	466.25	245	15.46	7384

Assumptions, parameters chosen unknown. Dates unknown.

Figure 2 Estimated failure rate results from a manufacturer.

The middle column labeled "Field Proven MTBF (Yrs.);" shows the manufacturer's estimated failure data in units of Mean Time Before Failures (MTBF). The column to the right shows the equivalent result in FITS (failures per billion hours). No methods/assumptions were given in the report from which this chart was copied. But these estimated results are likely based upon the manufacturer's warranty return system. It is likely that this is a good example of optimistic assumptions regarding the number of failures and the operating time.

Good quality field failure data needed for failure rate estimation is scarce but available. Failure rate estimations from this data can be quite useful. Many consider the OREDA database results [6,7] as the best in the world.

Along with the bad assumptions problem, another fundamental problem with field failure estimation is the time lag. **Given the pace of technology and quality improvement, high quality field failure data is often not available until the technology in use is obsolete.**

Failure Rate Prediction

To overcome the severe data time lag of estimation, **predictive methods** were developed starting in the late 1950s [8] in order to predict device failure rates for new designs. The intent was to analyze the components used, the design strength, and the stress conditions in order to predict a failure rate that will match the field failure results. Figure 3 shows the same chart as in Figure 2 with failure rate predictions using MIL-HNBK-217 [9] according to the report from which the chart was copied.

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Figure 3 Example of manufacturer supplied failure rate calculations using MIL-HDBK-217F.

The column labeled “Predicted MTBF” gives the results in MTBF. The column to the right gives the results in FITS. Obviously, the Figure 3 failure rates predicted do not even come close to matching the estimated failure rates from Figure 2 for the same products. Therefore, one can conclude that the prediction method used is not correct for these products. **When estimated failure rates do not match predicted failures rates, one or both methods have a serious problem. Such results cannot be considered realistic or even valid. Failure rate predictions must match quality failure rate estimations to be considered realistic and useable (Figure 4).**

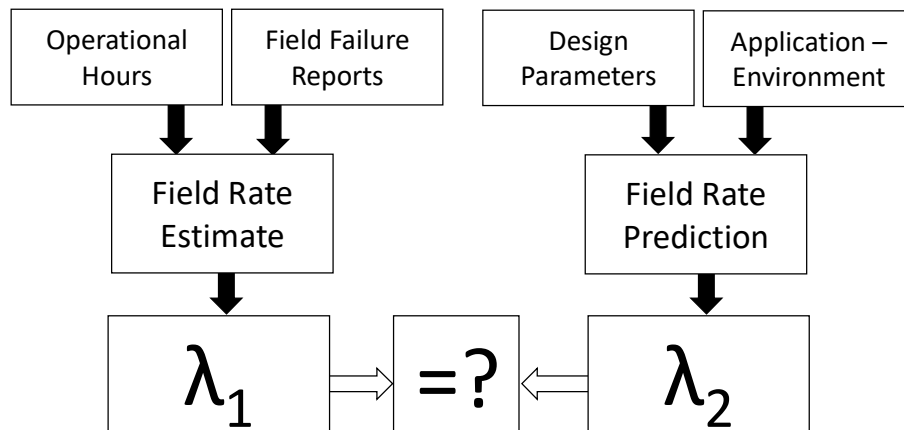


Figure 4 Estimated failure rates versus predicted failure rates.

Experienced reliability engineers want realistic data based on high quality field failure data. IEC 61511:2016 [10], Clause 11.9.3 states “The reliability data used when quantifying the effect of random failures shall be credible, traceable, documented, justified and shall be based on field feedback from similar devices used in a similar operating environment.” It is possible to get realistic data from either estimation or prediction. However, the most reliable data is the result of a combination of both methods where a prediction method is validated by estimated field failure data.

Quality Failure Rate Estimation

Failure rates can be estimated from failure reports and these estimates can be accurate. This can occur when the person performing the analysis has:

- a *complete set of information*,
- a good understanding of the equipment operation,
- quality procedures for data collection,
- a clear understanding of the application and environmental variables that impact failure,
- and a complete picture of how the equipment is used.

The three common sources of failure rate estimations are the manufacturer, industry databases, and site-specific data collection by device end user companies. Each source has advantages and disadvantages.

Manufacturer’s Field Return Data Studies

As seen in the charts above, a common source of failure data for estimation is manufacturer’s field return data. However, often assumptions must be made as there are several variables that are rarely known by the manufacturer, including the exact percentage of failed units that are returned and the actual operating hours. When assumptions are made optimistically, very low failure rates result. With realistic assumptions stated as upper and lower bounds, manufacturer field return data collection can be valuable to set upper and lower bounds on failure rates. And when the manufacturer identifies root causes of the failures, the data is very useful for verification of component failure rates and modes.

Industry Database

Failure rate data can also be found in industry failure databases and/or data books. These failure records are gathered by expert analysts, who make operational hour estimates, examine failure reports, and calculate failure rates. The resulting analysis is often published in a book or provided in a database.



One of the highest quality failure rate databases in the process industries is the OREDA database [6, 7]. Failure reports are submitted by member companies for their offshore operations in the North Sea. Data books present failure rate data for many types of process equipment.

Failures initiated by humans are included in the OREDA failure rate estimates [11]. Since this data collection system records all failures, random and systematic, the estimated failure rates include not only the inherent device failures but also failures due to site specific end-user actions. This is an excellent approach for obtaining realistic data.

The OREDA databases are consistently high quality, however the analysis only pertains to a certain assembly of equipment. Boundaries are defined for each assembly. Those boundaries are used to report data. The problem is that the lowest level assembly often includes more than one device, making it hard to tell which device was the cause of the failure. Therefore, the data are not available for a specific device in the published book. However, much information is made available from the OREDA analysts making the analysis methods quite transparent and useful.

Another industry database is the PERD, "Process Equipment Reliability Database," initiative from AIChE CCPS. This database initiative was created to provide device classifications, a consistent work process for data collection, and software tools [12].

Site Specific End User Field Failure Data Studies

Given the benefits of site collected failure data, many companies are setting up such programs. However, such efforts must be performed correctly to be of value. Some data collection systems have problems which result in very different failure rate results that are sometimes overly optimistic [13]. There are many possible reasons why different data collection systems provide varying estimated failure rates for the same product including scope of inclusions, environmental stressors, recognition systems, differences in site safety culture including proper policies and sufficiently trained site employees. Each one of these variations could change the resulting failure rate estimate by an order of magnitude or more. Therefore, the details of the data collection process must also be evaluated and understood by experts to validate a set of field failure data.

Analysis of some data collection systems have shown these problems. When the failure data collection process was examined, it became clear that systematic failures were not included in the failure count. This produced an unrealistically low failure rate. All failures should be recorded in a data collection system. Fortunately, several software tools [14, 15] have been created to accomplish this important data collection task and are able to collect needed details of each failure including the device's details comprising of the manufacturer's part and serial numbers, date of installation and current hours in operation. The system should also collect the physical environment at that time and the activity where the failure was recorded whether it be during an inspection, proof test, bypass or regular operation. The data collection system should



be easy to use with minimal or no added work for maintenance personnel and require less man-hours than any current methods

Failure Rate Prediction Methods

One problem with all failure rate estimation techniques is obsolescence. By the time enough data is gathered about a product, it may be obsolete, replaced by a new design which is not covered. Failure rate prediction methods were developed to overcome this issue. These are methods that may be used on newly designed devices to predict the failure rates and failure modes of the device. Two approaches have become well known, B10 cycle testing and FMEDA.

B10 Cycle Testing

Mechanical and electro-mechanical devices can wear out due to mechanical movement. The wear-out failure point is measured by the number of cycles performed before the failure. This is determined by "cycle testing [3]." In this testing, a set of devices are cycled until 10% of the units fail. The average number of cycles successfully completed is called the B10 point. The B10 point is converted to a time period by knowing the cycles per hour in any particular application. A failure rate is calculated by dividing the 10% failure count by the time period. A safe/dangerous failure mode split is arbitrarily assumed to be 50/50.

There have been ongoing debates in the reliability engineering community about the relationship between useful life failure rate and the wear-out point. There is some logic that supports the assumption of a longer useful life corresponding to a lower random failure rate although this theory is not supported by published research. The supporting assumption is that all failures during the useful life of a device are due to premature wear-out and that no other failure modes exist, such as corrosion binding, cold welding, erosion, stiction, and environmental stress. Therefore, the B10 method cannot be used if any of these other failure mechanisms are significant and if the device does not have mechanical movement at least once per week.

Failure Modes Effects and Diagnostic Analysis (FMEDA)

FMEDA is a predictive failure rate method based on a detailed study of design strength versus operational profile stress. The method was developed by engineers at exida in the late 1980s. It combines the decades old FMEA analysis approach with methods for calculating the diagnostic coverage factor (of automatic diagnostics and manual tests) [16], and safety system failure modes [17]. The FMEDA method requires a study of each component and how the component failure will affect the device.

A FMEDA (Figure 5) is done by examining each component in a device, one at a time. The stress conditions for the component are estimated or measured. For each failure mode of each component, the failure rate and effect on the product is recorded. The method is detailed and

systematic. With an understanding of fundamental failure modes in relation to specific applications, the failures can be classified as fail dangerous, fail safe, or other categories.

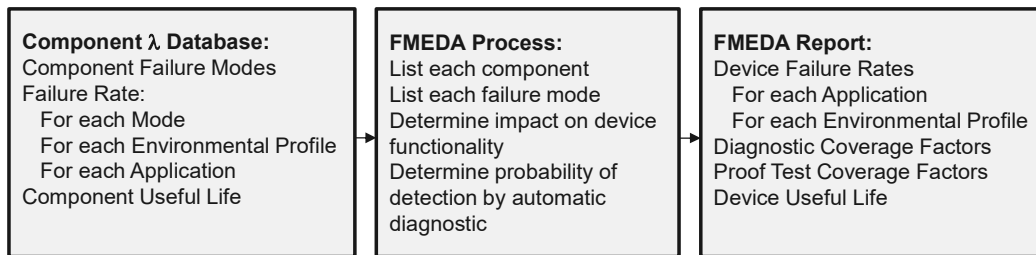


Figure 5. FMEDA Process

The FMEDA considers each component in the design including its design strength, functionality, failure modes and useful life. It also considers the operational profile including environmental stress factors and the impact a component failure will have on the product functionality. Also considered are the ability to detect a failure through automatic diagnostics or any proposed manual proof test.

A component failure rate is obtained as a function of operating environment. That component failure rate is entered in the FMEDA table according to the device failure mode. The probability that an automatic diagnostic will detect the specific failure mode is entered and the failure rates for each failure mode are calculated. The result is a product specific, application specific set of failure data that includes failure rates for each failure mode, failure rates that are detected and undetected by diagnostics, diagnostic coverage factors, useful life, and any proposed proof test coverage factors.

Calibrated FMEDA™ – Combining Estimation and Prediction

The FMEDA method can predict realistic application failure rates including human induced failures if the component failure rate database is calibrated from application field failure data [18]. This type of FMEDA from exida is called a “Calibrated FMEDA™”. Figure 6 shows a closed loop comparison method of component data calibration. As the quality and quantity of field failure data improve, the comparisons are repeated. After several years of field failure research and comparison, the FMEDA prediction produces ever more realistic results [19].

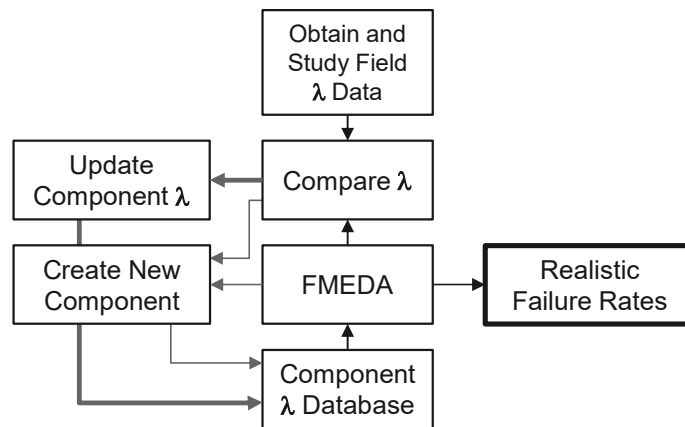


Figure 6. Field failure calibration of FMEDA component database.

A comparison of failure rates

An essential step in validation of the component database used for FMEDA is the repeated comparison of FMEDA predictions to actual field failure data. This method combines the best characteristics of estimation and prediction. Failure rates obtained from industry databases, manufacturer field failure studies, company failure records or other sources can be compared with FMEDA results. Often a scatter diagram showing variation is useful as shown in Figure 7. Calibrated FMEDA results for total failure rates of 35 different pressure transmitters from the process industries are shown as small dots. Also shown in large circles are the OREDA data point [20] and the equivalent data point from a field failure study published by DOW engineers [21], and the FMEDA average data point. The FMEDA average is realistically close to the field failure data estimates although the FMEDA average might be considered pessimistic as it is slightly higher.

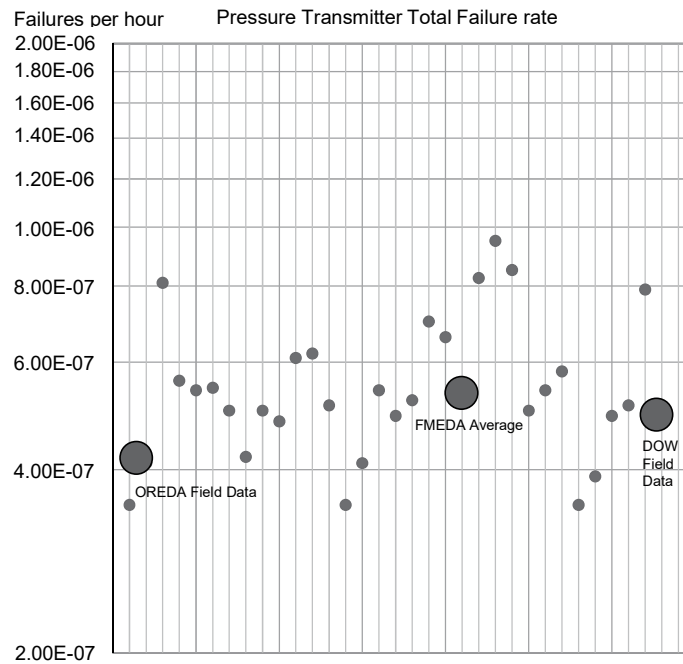


Figure 7 Pressure transmitter failure rate comparison of calibrated FMEDA results and field failure data.

Figure 8 shows a failure rate comparison of solenoid valves. Five calibrated FMEDA results are shown with field failure rate derived from OREDA data, estimated field failure data from a pipeline company, and published field failure data by DOW engineers. The line across the chart is the lower limit of a realistic process industry failure rate based on statistical analysis of the FMEDA data and field failure data. Any failure rate below this line must be carefully justified or preferably rejected. Also shown are published data points from other sources, using the B10 method [22, 23]. These numbers are far below the realistic lower limit [24].

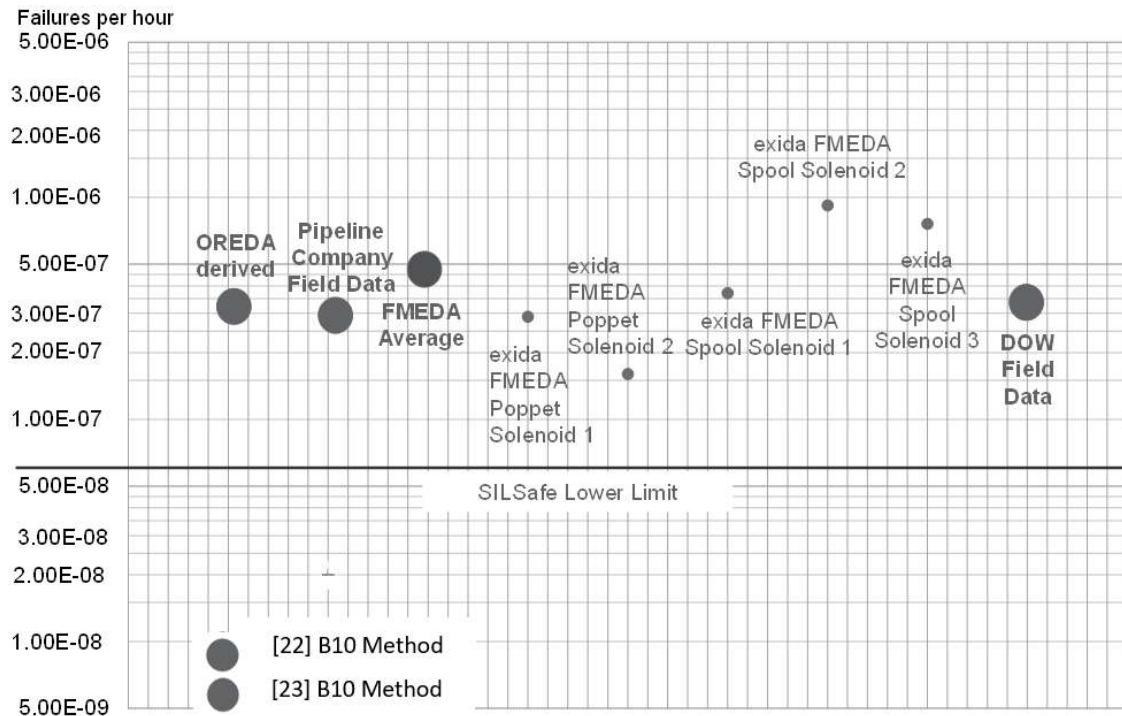


Figure 8 Solenoid valve failure rate comparison of calibrated FMEDA results, field failure data, and other manufacturer / certificate results.

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Revision History

Draft	Revision 0.1	January 8, 2018	W.M. Goble
Review/Release	Revision 1.0	June 1, 2018	W.M. Goble
Update w/ new references, updated info, charts			
	Revision 1.1	September 21, 2020	W.M. Goble
Update after feedback	Revision 1.2	September 23, 2020	W.M. Goble



exida – Who we are.

exida is one of the world's leading accredited certification and knowledge companies specializing in automation system cybersecurity, safety, and availability. Founded in 2000 by several of the world's top reliability and safety experts, exida is a global company with offices around the world. exida offers training, coaching, project-oriented consulting services, standalone and internet-based safety and cybersecurity engineering tools, detailed product assurance and certification analysis, and a collection of online safety, reliability, and cybersecurity resources. exida maintains a comprehensive failure rate and failure mode database on electrical and mechanical components, as well as automation equipment based on hundreds of field failure data sets representing over 350 billion unit operating hours.

exida Certification is an ANSI (American National Standards Institute) accredited independent certification organization that performs functional safety (IEC 61508 family of standards) and cybersecurity (IEC 62443 family of standards) certification assessments.

exida Engineering provides the users of automation systems with the knowledge to cost-effectively implement automation system cybersecurity, safety, and high availability solutions. The exida team will solve complex issues in the fields of functional safety, cybersecurity, and alarm management, like unique voting arrangement analysis, quantitative consequence analysis, or rare event likelihood analysis, and stands ready to assist when needed.

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exida believes that safety, high availability, and cybersecurity are achieved when more people understand the topics. Therefore, exida has developed a successful training suite of online, on-demand, and web-based instructor-led courses and on-site training provided either as part of a project or by standard courses. The course content and subjects range from introductory to advanced. The exida website lists the continuous range of courses offered around the world.

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Tools and Products for End User Support

- exSiLentia® – Integrated Safety Lifecycle Tool
 - PHAx™ (Process Hazard Analysis)
 - LOPAx™ (Layer of Protection Analysis)



excellence in dependable automation

- SILAlarm™ (Alarm Management and Rationalization)
- SILect™ (SIL Selection and Layer of Protection Analysis)
- Process SRS (PHA based Safety Requirements Specification definition)
- SILver™ (SIL verification)
- Design SRS (Conceptual Design based Safety Requirements Specification definition)
- Cost (Lifecycle Cost Estimator and Cost Benefit Analysis)
- PTG (Proof Test Generator)
- SILstat™ (Life Event Recording and Monitoring)
- exSILentia® Cyber- Integrated Cybersecurity Lifecycle Tool
 - CyberPHAx™ (Cybersecurity Vulnerability and Risk Assessment)
 - CyberSL™ (Cyber Security Level Verification)

Tools and Products for Manufacturer Support

- FMEDAx (FMEDA tool including the exida EMCRH database)
- ARCHx (System Analysis tool; Hardware and Software Failure, Dependent Failure, and Cyber Threat Analysis)

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